Mayank Malhotra

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.28
DELHI PUBLIC SCHOOL ALLAHABAD	Class XII (CENTRAL BOARD OF SECONDARY EDUCATION)	2021	93.60%
DELHI PUBLIC SCHOOL ALLAHABAD	Class X (CENTRAL BOARD OF SECONDARY EDUCATION)	2019	91.83%

PROJECTS

GST Audit Review & Legal RAG Platform

- Built an hybrid RAG platform over **114K+ GST statutory chunks** and **17.4K+ judicial rulings**, scoring **94.0% Recall@5, 75.9% Recall@1 ,** and **0.828 MRR.**
- Fused **SQLite FTS5/BM25 + Qdrant HNSW (bge-m3)** using Reciprocal Rank Fusion and **bge-reranker-v2-m3**, improving **Recall@1 from 64.7% to 75.9%** with **sub-50ms retrieval latency**.
- Served **Qwen3.6-27B FP8 via vLLM on H100 NVL**, supporting **256 concurrent sequences** and **48-way parallel enrichment** while keeping the KV cache within VRAM.

Agentic Customer Support System with LLM Distillation

- Built a **tool-using AI agent** for end-to-end e-commerce support ticket resolution (orders, tracking, refunds, cancellations, escalation), with **policy guardrails enforced in code.**
- **Distilled a 122B agent into a LoRA fine-tuned 3B model** using only correct teacher trajectories; on 186 unseen adversarial tickets, it improved resolution from 63% to 86%, matching the teacher, while reducing GPU serving from **4 H200 GPUs to 1 A100 GPU**, with **40 times smaller weights** (234 GB to 5.8 GB) and **25% lower latency**.

Secure Agentic Analytics System (Text-to-SQL)

- Built an agentic analytics system over a TPC-DS warehouse (25 tables, 2.9M sales rows) that discovers tables, applies **certified metric definitions**, and asks clarifying questions, lifting correct answers on ambiguous metrics from **45% to 100%**; traced **16/20 deliberately injected revenue anomalies** to their category and region.
- Enforced **role-based PII governance in code** via scope-aware SQL parsing across joins, CTEs, aliases, and SELECT *: **0 leaks across 91 adversarial attacks** with no false refusals on 35 legitimate queries, while prompt-only rules on a 14B model were bypassed by 1 of 4 prompt-injection techniques.
- Reached **65.3% execution accuracy on BIRD development set** (official evaluator, 1,534 questions).

Raft-KV: Distributed Key-Value Store with Consensus from Scratch

- Implemented **Raft consensus from scratch** (leader election, log replication, snapshots, and log compaction) over gRPC; 3 nodes sustained **8.6K quorum-committed writes/s** and **9.7K linearizable ReadIndex reads/s** at **3.5ms p50**.
- Built an automated **fault-tolerance test suite**: leader failure triggered re-election in **327-486ms** with zero committed-data loss; followers with compacted logs rejoined via snapshots, and restarts recovered state from disk.

QR Pass & Token Management System

- Replaced paper tickets with a **QR pass system** for Prakshepan 4.0, generating a unique pass for every registered attendee and enabling volunteers to check people in by scanning passes with a phone.
- Designed passes with **entry limits** so each code works only for its allowed number of entries, preventing reuse by multiple people and eliminating manual list-checking at the gate.

PUBLICATION

Leveraging BERT and GPT for Enhanced Fake News Detection and Analysis

- IEEE, 2025 5th International Conference on Soft Computing for Security Applications (ICSCSA), pp. 1341-1347 DOI:10.1109/ICSCSA66339.2025.11171041
- Developed and evaluated **BERT and GPT based models for fake news detection and comparative analysis**.

TECHNICAL SKILLS

Languages: Python, C++, C, JavaScript, SQL

AI/ML: GenAI, LLMs, RAG, Agentic AI, Fine-Tuning, Knowledge Distillation, Quantization, Inference Optimization, Embeddings, Retrieval & Reranking, Model Evaluation, NLP, Transformers

Frameworks & Libraries: PyTorch, FastAPI, React, Next.js, Node.js, TensorFlow, Scikit-learn

Systems & Tools: Docker, Slurm, vLLM, Qdrant, SQLite, Git, GitHub, REST APIs

POSITIONS OF RESPONSIBILITY

Teaching Assistant, IIT Hyderabad -Fundamentals of Machine Learning(CS 5590/3390)

Teaching Assistant, IIT Hyderabad -Theory of Computation(CS 2030)

Teaching Assistant, IIT Hyderabad -Data Structures & Algorithms(ID 2230)

Core Member, Prakshepan 4.0 Organizing Team, IIT Hyderabad

RELEVANT COURSES

Relevant Courses:

- **AI/ML:** Fundamentals of Machine Learning, Natural Language Processing, Deep Learning
- **Systems & Algorithms:** Advanced Data Structures & Algorithms, Advanced Computer Networks